Linköping University | Department of Industrial Engineering
Master's thesis, 30 ECTS |
2026 | LIU-IEI/LITH-EX-A--26/001--SE

Skewness and Concentration in Nordic Equity Markets: Implications for Portfolio Construction

Working on it

Leo Jozic
William Öbrink

Supervisor : Taras Bodnar
Examiner : Jörgen Blomvall

LINKÖPING UNIVERSITY

Linköpings universitet
SE–581 83 Linköping
+46 13 28 10 00 , www.liu.se

Upphovsrätt

Detta dokument hålls tillgängligt på Internet - eller dess framtida ersättare - under 25 år från publiceringsdatum under förutsättning att inga extraordinära omständigheter uppstår.

Tillgång till dokumentet innebär tillstånd för var och en att läsa, ladda ner, skriva ut enstaka kopior för enskilt bruk och att använda det oförändrat för ickekommersiell forskning och för undervisning. Överföring av upphovsrätten vid en senare tidpunkt kan inte upphäva detta tillstånd. All annan användning av dokumentet kräver upphovsmannens medgivande. För att garantera äktheten, säkerheten och tillgängligheten finns lösningar av teknisk och administrativ art.

Upphovsmannens ideella rätt innefattar rätt att bli nämnd som upphovsman i den omfattning som god sed kräver vid användning av dokumentet på ovan beskrivna sätt samt skydd mot att dokumentet ändras eller presenteras i sådan form eller i sådant sammanhang som är kränkande för upphovsmannens litterära eller konstnärliga anseende eller egenart.

För ytterligare information om Linköping University Electronic Press se förlagets hemsida `http://www.ep.liu.se/`.

Copyright

The publishers will keep this document online on the Internet - or its possible replacement - for a period of 25 years starting from the date of publication barring exceptional circumstances.

The online availability of the document implies permanent permission for anyone to read, to download, or to print out single copies for his/hers own use and to use it unchanged for non-commercial research and educational purpose. Subsequent transfers of copyright cannot revoke this permission. All other uses of the document are conditional upon the consent of the copyright owner. The publisher has taken technical and administrative measures to assure authenticity, security and accessibility.

According to intellectual property law the author has the right to be mentioned when his/her work is accessed as described above and to be protected against infringement.

For additional information about the Linköping University Electronic Press and its procedures for publication and for assurance of document integrity, please refer to its www home page: `http://www.ep.liu.se/`.

© Leo Jozic
William Öbrink

Abstract

The abstract resides in file `Abstract.tex`. Here you should write a short summary of your work.

Lorem ipsum dolor sit amet, consectetur adipiscing elit. Pellentesque in massa suscipit, congue massa in, pharetra lacus. Donec nec felis tempor, suscipit metus molestie, consectetur orci. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Curabitur fermentum, augue non ullamcorper tempus, ex urna suscipit lorem, eu consectetur ligula orci quis ex. Phasellus imperdiet dolor at luctus tempor. Curabitur nisi enim, porta ut gravida nec, feugiat fermentum purus. Donec hendrerit justo metus. In ultrices malesuada erat id scelerisque. Sed sapien nisi, feugiat in ligula vitae, condimentum accumsan nisi. Nunc sit amet est leo. Quisque hendrerit, libero ut viverra aliquet, neque mi vestibulum mauris, a tincidunt nulla lacus vitae nunc. Cras eros ex, tincidunt ac porta et, vulputate ut lectus. Curabitur ultricies faucibus turpis, ac placerat sem sollicitudin at. Ut libero odio, eleifend in urna non, varius imperdiet diam. Aenean lacinia dapibus mauris. Sed posuere imperdiet ipsum a fermentum.

Nulla lobortis enim ac magna rhoncus, nec condimentum erat aliquam. Nullam laoreet interdum lacus, ac rutrum eros dictum vel. Cras lobortis egestas lectus, id varius turpis rhoncus et. Nam vitae auctor ligula, et fermentum turpis. Morbi neque tellus, dignissim a cursus sed, tempus eu sapien. Morbi volutpat convallis mauris, a euismod dui egestas sit amet. Nullam a volutpat mauris. Fusce sed ipsum lectus. In feugiat, velit eu fermentum efficitur, mi ex eleifend ante, eget scelerisque sem turpis nec augue.

Vestibulum posuere nibh ut iaculis semper. Ut diam justo, interdum quis felis ac, posuere fermentum ex. Fusce tincidunt vel nunc non semper. Sed ultrices suscipit dui, vel lacinia lorem euismod quis. Etiam pellentesque vitae sem eu bibendum. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas. Pellentesque scelerisque congue ullamcorper. Sed vehicula sodales velit a scelerisque. Pellentesque dignissim lectus ipsum, quis consectetur tellus rhoncus a.

Nunc placerat ut lectus vel ornare. Sed nec dictum enim. Donec imperdiet, ipsum ut facilisis blandit, lacus nisi maximus ex, sed semper nisl metus eget leo. Nunc efficitur risus ac risus placerat, vel ullamcorper felis interdum. Class aptent taciti sociosqu ad litora torquent per conubia nostra, per inceptos himenaeos. Duis vitae felis vel nibh sodales fringilla. Donec semper eleifend sem quis ornare. Proin et leo ut dolor consectetur vehicula. Lorem ipsum dolor sit amet, consectetur adipiscing elit.

Nunc dignissim interdum orci, sit amet pretium nibh consectetur sagittis. Aenean a eros id risus aliquam placerat nec ut lectus. Curabitur at quam in nisi sodales imperdiet in at erat. Praesent euismod pulvinar imperdiet. Nam auctor mattis nisi in efficitur. Quisque non cursus ipsum, consequat vehicula justo. Fusce varius metus et nulla rutrum scelerisque. Praesent molestie elementum nulla a consequat. In at facilisis nisi, convallis molestie sapien. Cras id ullamcorper purus. Sed at lectus sit amet dolor finibus suscipit vel et purus. Sed odio ipsum, dictum vel justo sit amet, interdum dictum justo. Quisque euismod quam magna, at dignissim eros varius in. Pellentesque habitant morbi tristique senectus et netus et malesuada fames ac turpis egestas.

Acknowledgments

Acknowledgments.tex

Contents

Abstract iii

Acknowledgments iv

Contents v

List of Figures vii

1 Introduction 2
- 1.1 Kvartil . . . 3
- 1.2 Purpose . . . 3
- 1.3 Research questions . . . 3
- 1.4 Delimitations . . . 3

2 Background 5
- 2.1 Return distribution on financial markets . . . 5
- 2.2 Skewness in return distribution . . . 6
- 2.3 Long-term stock return and concentration . . . 7
- 2.4 Implications for portfolio management . . . 8

3 Methodology 9
- 3.1 Literature study . . . 10
- 3.2 Data collection . . . 10
- 3.3 Data processing . . . 11
- 3.4 Modeling . . . 12
- 3.5 Evaluation . . . 13

4 Theory 14
- 4.1 Arithmetic returns . . . 14
- 4.2 Logarithmic returns . . . 15
- 4.3 Skewness . . . 16
- 4.4 Wealth creation . . . 16
- 4.5 Statistical t-test . . . 17
- 4.6 Monte carlo simulation . . . 19

5 Method 20

6 Results 21

7 Discussion 22
- 7.1 Results . . . 22
- 7.2 Method . . . 22
- 7.3 The work in a wider context . . . 23

8 Conclusion **24**

Bibliography **25**

List of Figures

2.1 Illustration of different skewed distributions. Source: (Wikimedia Commons contributors, n.d.). 7

3.1 Overview of the methodology of the study. 10

Nomenclature

1 Introduction

Research conducted in U.S. equity markets shows that the majority of buy-and-hold common stock returns are outperformed by returns on one-month U.S. Treasury bills held over the same investment horizon (Bessembinder, 2018). This finding challenges the conventional view of equity markets as broadly rewarding for long-term investors and suggests that aggregate market returns may obscure substantial heterogeneity in individual stock outcomes.

Equity markets play a central role in long-term wealth accumulation, pension systems, and capital allocation. Market performance is therefore often summarized using long-run average returns or broad market indices. While such measures provide a useful aggregate perspective, they offer limited insight into how returns are generated at the individual stock level. Aggregate market returns in the U.S. mask a highly uneven distribution of outcomes across firms, where a small number of stocks generate large positive returns while the majority deliver modest or negative performance over long horizons (Bessembinder, 2018).

Previous research has documented this pattern extensively for the U.S. equity market. In particular, Professor Hendrik Bessembinder shows in his research (Bessembinder, 2018) that most individual stocks underperform certain risk-free assets over their lifetimes, while nearly all long-term net wealth creation is attributable to a small fraction of firms. These results imply that individual stock returns are highly right-skewed and that aggregate market performance is driven by a limited number of extreme winners.

However, comparable evidence for Nordic equity markets is limited. This is notable given that Nordic markets differ from the U.S. in terms of market size, sector composition, ownership structures and institutional characteristics. It is therefore of interest to examine whether a similar return structure is present in Nordic equity markets or whether these structural differences are associated with different patterns of return skewness and concentration.

Understanding the distribution of individual stock returns has important implications for portfolio construction and active asset management. In the presence of highly skewed return distributions, diversification primarily serves to increase the probability of capturing extreme positive outcomes rather than merely reducing volatility (Reinholtz, Fernbach, and Langhe, 2016). Similarly, the performance of concentrated portfolios and stock-picking strategies

must be assessed in a setting where underperformance relative to the market is the most likely outcome, despite the existence of a small number of highly successful investments.

By combining an empirical analysis of individual stock return distributions, a wealth-creation framework, and a portfolio-level evaluation, this study contributes to the literature by extending existing evidence beyond the U.S. market and by providing insights into the consequences of skewed return distributions for long-term investors in Nordic equity markets.

1.1 Kvartil

This thesis is conducted in collaboration with Kvartil, an asset management firm founded in 2018 that combines data, statistical analysis, and systematic rules in its investment process to enhance decision-making and portfolio construction. Kvartil was founded by former employees of Brummer Partners, bringing extensive experience from quantitative asset management. The team at Kvartil has contributed with domain expertise, valuable insights, and guidance throughout this thesis.

1.2 Purpose

The purpose of this study is to examine the achieved returns from portfolios constructed with different methods in a market environment characterized by positively skewed and concentrated individual stock returns.

1.3 Research questions

1. Do pooled return distributions in Nordic equity markets display significant positive skewness across different investment horizons?
2. How concentrated is long-term wealth creation across firms in the Nordic equity markets?
3. How do different portfolio construction methods (including portfolio size and random and informed selection strategies) affect portfolio return distributions in a market characterized by skewed and concentrated stock returns?

1.4 Delimitations

To maintain a clear and well-defined scope, and to ensure consistency and availability of the data, this thesis applies several delimitations to the study design.

Nordic markets
This study is exclusively focused on the Nordic equity markets. By limiting the scope to the Nordic markets, the analysis aims to provide a more internally consistent setting in which observed return patterns can be attributed to underlying market dynamics rather than cross-market structural differences. The following markets are the ones considered in this study:

- Sweden
  - Nasdaq Stockholm Stock Exchange
  - Spotlight Stock Market

- Denmark
  - Nasdaq Copenhagen Stock Exchange
- Norway
  - Euronext Oslo Stock Exchange
- Finland
  - Nasdaq Helsinki Stock Exchange
- International
  - Nordic Growth Market

Time horizon and return periods
The analysis considers one-year, five-year, and lifetime return horizons in order to capture both short- to long-term return dynamics and long-run wealth creation. For the one-year and five-year horizons, the sample is restricted to the most recent 35-year period. This choice reflects a trade-off between obtaining a sufficiently large number of observations and maintaining relevance to the current and recent market environment.

Data selection
For the one-year and five-year return analyses, the dataset includes return observations from periods during which companies were publicly listed and actively traded on the Nordic equity markets. In contrast, the lifetime return analysis includes all companies that have been listed on the Nordic markets at some point in the sample time, covering their entire listing history from initial listing to delisting or the end of the sample period.

2 Background

This chapter provides the background necessary to understand the theoretical aspects of stock returns and their implications for investment outcomes. It begins by introducing the concept of return distributions and the statistical characteristics commonly observed in financial data. Particular attention is given to skewness, which reflects the asymmetric nature of stock returns and highlights the presence of extreme outcomes. The chapter then examines how such asymmetries translate into concentration in long-term market performance, where a limited number of firms account for a disproportionate share of total wealth creation. The concept of wealth creation is introduced as a framework for quantifying the economic contribution of individual firms to aggregate market outcomes. Finally, the chapter discusses the implications of the empirically observed skewness in pooled return observations, together with the concentration of long-term wealth creation, for portfolio construction. This provides the conceptual foundation for the analysis that follows, which investigates how different portfolio construction approaches perform in an environment characterized by asymmetric and highly concentrated return outcomes.

2.1 Return distribution on financial markets

In financial markets, asset returns are commonly analyzed through their statistical distributions. A return distribution describes how frequently different return outcomes occur over a given horizon and provides a compact representation of both typical and extreme outcomes. Understanding the shape of return distributions is therefore a fundamental prerequisite for analyzing risk, performance, and investment outcomes.

In empirical studies, returns are typically measured as arithmetic or logarithmic price changes over fixed intervals, such as daily, monthly, or annual horizons (Miskolczi, 2017). The collection of these observations over time forms an empirical return distribution for a specific firm, which can be summarized using descriptive statistics such as the mean, median, skewness, and standard deviation. While these summary measures provide useful information, they do not fully characterize the distributional shape of returns.

In much of classical financial theory, asset prices are commonly modeled as following a geometric Brownian motion. Under this assumption, continuously compounded logarithmic

returns are normally distributed, while prices themselves follow a lognormal process (Mota, 2012). However, a large amount of empirical evidence suggests that observed logarithmic return distributions frequently deviate from normality. Observed return distributions often exhibit features such as asymmetry, heavy tails, and excess kurtosis, meaning that extreme outcomes occur more frequently than predicted by a normal distribution (Cont, 2001). These deviations are observed across asset classes, markets, and time periods, and become particularly pronounced when returns are measured over longer horizons or at the level of individual stocks.

As a result, empirical return distributions are typically analyzed using non-parametric tools, such as histograms, which allows one to visualize the data without imposing strong distributional assumptions. These tools provide a flexible way to visualize and summarize return behavior and serve as a natural starting point for further distributional analysis.

This study adopts such an empirical perspective when examining stock returns in Nordic equity markets. Rather than assuming a particular parametric form, the analysis begins by documenting the observed distributional properties of returns across different horizons. This approach allows for a transparent assessment of how closely empirical return distributions align with standard theoretical assumptions and provides the foundation for subsequent analysis of asymmetry, concentration, and portfolio-level implications.

2.2 Skewness in return distribution

Skewness is a fundamental property of return distributions that captures the degree of asymmetry around the central tendency. In a positively skewed distribution, the right tail is longer or heavier than the left tail, implying that a small number of observations take on exceptionally large positive values, while the majority of observations are clustered around lower or moderate outcomes. (Doane and Seward, 2011)

In the context of financial returns, when individual stocks are considered as separate observations, their long-term return outcomes exhibit a highly positively skewed cross-sectional distribution. Empirically, the distribution of individual U.S stock returns is characterized by many small losses or modest gains, combined with a small number of extreme winners (Bessembinder, 2018). As a result, the distribution of returns is asymmetric, and measures of central tendency such as the mean and the median can differ substantially. The distinction between mean and median outcomes is economically important. In a positively skewed distribution, the mean is pulled upward by rare but extreme positive outcomes, while the median reflects the outcome experienced by a typical observation (Doane and Seward, 2011). This feature implies that overall market performance may be driven by a limited number of exceptional firms rather than reflecting the typical outcome across stocks.

Recent empirical evidence highlights the extent to which individual stock returns exhibit such skewed behavior. Bessembinder (2018) documents that, over long horizons, the majority of U.S. stocks generate returns below those of one month Treasury bills, while a relatively small subset of firms accounts for nearly all positive net wealth creation in the market. These findings illustrate that aggregate market performance is driven by a small number of extreme winners, while the typical stock contributes little to long-term wealth creation.

Importantly, skewness is a defining feature of equity return distributions with direct economic implications. When returns are highly skewed, investment outcomes become sensitive to the inclusion or exclusion of a small number of extreme winners. This property challenges

intuitive interpretations based on average performance and motivates a closer examination of how skewed stock-level returns aggregate at the market and portfolio level.

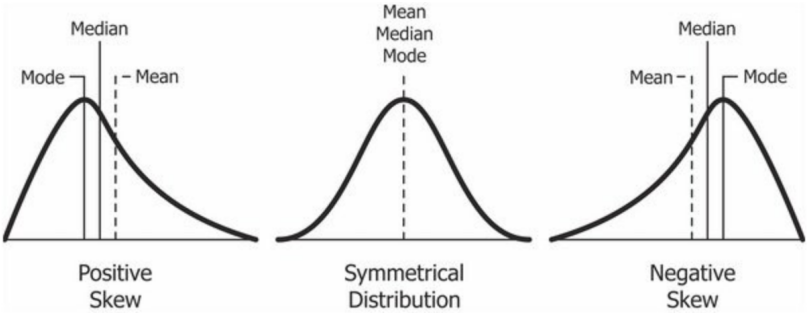

Figure 2.1: Illustration of different skewed distributions. Source: (Wikimedia Commons contributors, n.d.).

2.3 Long-term stock return and concentration

While skewness describes asymmetry in return distributions, it does not fully capture the economic importance of extreme outcomes. To assess their market-level relevance, it is necessary to examine how long-term stock returns and wealth creation are distributed across firms. When returns are measured over long horizons, such as several years or over a firm's entire publicly traded lifetime, each stock contributes a single cumulative outcome. Empirical evidence shows that this distribution is highly uneven, where most stocks generate modest or negative long-term returns, while a small number of firms achieve exceptionally large gains (Bessembinder, 2018).

This uneven contribution is captured by the concept of concentration, which refers to how total market wealth creation is distributed across firms. Aggregate market performance may therefore be driven by a limited subset of companies, even when the overall market exhibits strong long-term returns. Bessembinder (2018) formalizes this idea using the concept of wealth creation, defined as the cumulative value generated by a firm's equity in excess of a risk-free investment over its lifetime. Using this measure, he documents that nearly all net wealth creation in the U.S. equity market can be attributed to a small fraction of firms, while the typical stock underperforms risk-free assets (Bessembinder, 2018).

Applying the wealth creation framework further enables an investigation of the temporal structure of skewness. Because wealth creation accumulates multiplicatively over time, it offers a direct way to examine whether return concentration increases with the investment horizon. This allows the study to distinguish between skewness arising from the cross-sectional distribution of single-period returns and skewness that is amplified through long-term compounding effects.

Wealth creation thus provides a direct link between skewness at the individual stock level and concentration in aggregate market outcomes. This perspective motivates an examination of whether similar patterns of concentration are present in Nordic equity markets and sets the stage for analyzing their implications for portfolio-level outcomes.

2.4 Implications for portfolio management

The presence of skewness and concentration in the return distribution have an impact when it comes to portfolio construction. When the long-term returns are unevenly distributed across firms, aggregate market performance is driven by a relatively small number of extreme positive outcomes. As a result, investment outcomes depend critically on whether a portfolio includes the firms responsible for the majority of long-term value creation.

In such an environment, the interpretation of average returns becomes less informative for individual investors. While the mean return across stocks may be positive, due to the positive skewness in the return distribution, the typical stock may deliver substantially lower performance. This implies that investors holding only a limited number of securities face a significant probability of underperforming the broader market, even over long investment horizons. These characteristics highlight the role of diversification as a mechanism for increasing the likelihood of capturing the small subset of firms that generate the majority of long-term gains. Diversification, rather than solely reducing volatility (Reinholtz, Fernbach, and Langhe, 2016), improves the probability that a portfolio includes the extreme positive outcomes that drive aggregate market performance.

At the same time, different portfolio construction approaches may result in different exposure to these extreme outcomes. By sorting firms within size segments (small, mid and large cap) according to certain construction rules, and by varying the number of stocks included in each portfolio, the skewness exposure (TILLSAMMANS MED ANDRA REGLER) and diversification are analyzed to observe how they jointly shape portfolio return distributions in an environment characterized by concentrated wealth creation and an empirically skewed pooled distribution. This motivates an examination of how different portfolio construction rules affect the distribution of portfolio returns.

3 Methodology

This chapter outlines the methodological framework of the study. The objective is to examine whether Nordic equity markets exhibit skewed return distributions and concentrated wealth creation, and to analyze how these structural characteristics affect portfolio outcomes under different construction rules. The chapter begins with a review of relevant literature and a description of the data collection and processing procedures. It then presents the modeling framework, where skewness and wealth concentration are documented and portfolios are constructed based on predefined selection rules. Finally, an evaluation framework is introduced to assess differences between rule-based portfolios and to benchmark their performance against randomly formed portfolios. Together, these components provide a structured approach for linking stock-level return characteristics to portfolio-level performance in an environment characterized by asymmetric and concentrated return outcomes. All empirical analysis is conducted using data from LSEG Workspace and implemented in Python.

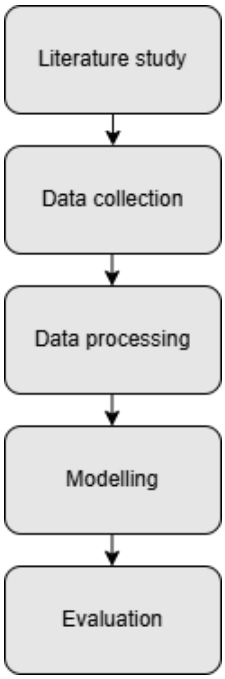

Figure 3.1: Overview of the methodology of the study.

3.1 Literature study

This study builds upon prior empirical research examining the distributional properties of individual stock returns and their implications for aggregate market outcomes. A contribution in this area is a study conducted by Bessembinder (2018), who analyzes the distribution of long-term stock returns using comprehensive data on U.S. equities. His research provides a framework for quantifying concentration in long-term equity returns and serves as a key reference point for this study. In addition to Bessembinders work, other research papers have been used that document that individual stock returns exhibit highly asymmetric distributions, characterized by positive skewness and the presence of extreme positive outcomes. This study builds on these approaches by applying similar distributional analyses to Nordic equity markets, specifically examining the empirical distribution of stock returns, quantify wealth creation and concentration, and construct rule-based portfolios to analyze how portfolio size and selection rules influence the distribution of portfolio outcomes.

3.2 Data collection

All necessary data for this study are obtained from LSEG Workspace.

(Måste kolla i refinitiv om det går att hämta data innan 1987)

3.2.1 Markets

To examine whether return skewness is present in Nordic equity markets, the analysis is based on firms listed on the major Nordic exchanges. Rather than focusing on a single country, the study covers multiple markets in order to capture the overall structure of return distributions in the Nordic region and to reduce the influence of country-specific effects.

The collected dataset includes all companies that were listed in any of the selected Nordic

equity markets at some point during the sample period. This implies that the sample is defined by listing location rather than country of incorporation. Consequently, a subset of the firms in the study may be registered or headquartered in another country while being publicly traded on a Nordic exchange.

From an investment perspective, what matters is where the security is traded and accessible to market participants, not necessarily where the firm is legally incorporated. By taking this approach the results describe the structure of returns in Nordic exchanges, rather than the performance of purely Nordic-domiciled firms. This distinction is important for interpretation, particularly when comparing findings to studies based on domestic firm samples in other regions.

The following markets are the ones included in the study:

- Sweden
  - Nasdaq Stockholm Stock Exchange
  - Spotlight Stock Market
- Denmark
  - Nasdaq Copenhagen Stock Exchange
- Norway
  - Euronext Oslo Stock Exchange
- Finland
  - Nasdaq Helsinki Stock Exchange
- International
  - Nordic Growth Market

3.2.2 Returns

This study is based on returns generated by publicly traded companies listed on the Nordic exchanges described in Section 3.2.1. For firms that were delisted during the sample period, the reason for delisting is identified and treated according to the defined data processing procedure 3.3.

In empirical finance and financial econometrics, returns are commonly expressed as logarithmic (continuously compounded) returns. This convention stems from their analytical convenience, particularly their time-additive property and their compatibility with geometric averaging and continuous compounding (Campbell, Lo, and MacKinlay, 1997). Accordingly, and in line with established practice, this study employs logarithmic returns throughout the analysis.

3.3 Data processing

To enable a consistent and reliable empirical analysis, the collected data were subject to a structured process of cleaning, processing, and formatting. The data preparation was conducted in Excel, including the use of built-in functions and VBA scripts to manage large datasets and automate repetitive procedures.

A central component of the data processing concerned the treatment of firms that were delisted during the sample period. For each delisted company, the reason for delisting was identified in order to ensure an economically correct measurement of the final return. In cases of bankruptcy or liquidation, the terminal return was recorded as a total loss, reflecting the fact that shareholders effectively lost their invested capital. In contrast, when a firm was delisted due to an acquisition or merger, the final observed return was calculated based on the last available price, including any compensation received by shareholders. This approach is similarly applied in (Bessembinder, 2018).

This distinction is crucial, as improper treatment of delistings may lead to survivorship bias (Financial Edge Training, 2022) in the distribution of returns, particularly in studies examining skewness and the concentration of long-term wealth creation.

3.4 Modeling

The modeling framework links stock-level return characteristics to portfolio-level outcomes. Specifically, the analysis proceeds in two stages. First, skewness and wealth concentration in individual stock returns are documented in order to establish the structural properties of the Nordic equity market. Second, portfolios could be constructed based on predefined selection rules to examine how these structural features translate into portfolio performance.

3.4.1 Observation of skewness and concentration among returns

The first step of this study is to conduct an empirical observation of the distributional properties of stock returns in the Nordic market and their implications for wealth creation. This empirical observation can be divided into two stages.

First, the distribution of stock return observations is examined across different investment horizons, including one-year, five-year, and lifetime returns. The analysis focuses on the empirical shape of these distributions using graphical methods such as histograms. By documenting asymmetry and tail behavior without imposing parametric assumptions, this step establishes whether return outcomes exhibit pronounced skewness and uneven dispersion.

Second, firm-level wealth creation is evaluated to assess the concentration of long-term wealth creation within the Nordic market. Wealth creation is defined as the cumulative value generated by a firm's change in market capitalization in excess of a risk-free investment over its publicly traded lifetime. Firms are then compared based on their relative contribution to total market wealth creation, allowing for an assessment of how unevenly wealth creation is distributed across firms.

3.4.2 Portfolio construction

To examine the implications of skewness in individual stock returns, a range of portfolios is constructed and analyzed. The objective is to evaluate differences in performance across portfolios and to examine how their return distributions vary depending on underlying stock characteristics.

HÄR KOMMER VALET AV PORTFÖLJER ATT BESKRIVAS. SKEWNESS FÖR INDIVIDUELLA FÖRETAG, DIVERSIFIERNING OCH STORLEK (CAP) KOMMER ATT VARA PARAMETRAR FÖR DETTA. VIDARE DISKUSSION MED KVARTIL FÖRS FÖR ATT TA ÖNSKEMÅL FÖR FLER PARAMETRAR/REGLER I BEAKTNING. TANKEN ÄR ATT ALLA

PORTFÖLJER ÄR EQUALLY WEIGHTED.

3.5 Evaluation

This section outlines the framework used to evaluate the performance of the constructed portfolios. The evaluation is conducted in two complementary stages. First, rule-based portfolios are compared statistically to assess whether systematic differences in construction rules translate into differences in average returns. Second, each rule-based portfolio is evaluated relative to a benchmark distribution of randomly formed portfolios in order to assess whether its performance is distinguishable from random selection.

3.5.1 Statistical comparison of rule-based portfolios

To compare portfolios constructed under different rules, daily return differences between two portfolios are computed over the evaluation period, which is a period of the most recent five years. The mean return difference is then tested for statistical significance.

The statistical significance of the estimated mean return differences depends critically on the relationship between the average return differential and its volatility. Since standard inference relies on the ratio between the mean and its standard error, which scales with the sample standard deviation and the square root of the number of observations, the amount of historical data required to detect statistically significant differences varies across portfolio pairs. In particular, when return differentials exhibit high volatility relative to their mean, longer time series are necessary to obtain sufficient statistical power.

Because daily financial returns may exhibit heteroskedasticity and serial dependence, inference is conducted using heteroskedasticity- and autocorrelation-consistent (HAC) standard errors (Newey and West, 1987). This approach provides robust standard errors that remain valid under general forms of time-series dependence. Statistical significance is assessed using the corresponding t-statistic and p-values derived from the asymptotic distribution of the estimator. The sign of the estimated mean difference determines which portfolio exhibits higher average performance, while statistical significance indicates whether the observed difference is unlikely to be driven by sampling variation.

3.5.2 Benchmarking against randomly formed portfolios

In addition to pairwise statistical tests, each rule-based portfolio is evaluated relative to a benchmark distribution of randomly formed portfolios. For each portfolio configuration, defined by size segment, diversification level, and (ANDRA REGLER), a large number of portfolios are generated through random selection of stocks under identical constraints. Furthermore, comparisons against the value-weighted market portfolio is made to provide a concrete benchmark that reflects aggregate market performance.

This Monte Carlo simulation approach produces an empirical distribution of portfolio outcomes that reflects the range of performance attainable through random diversification within the same investment universe. The realized performance of the rule-based portfolio is then evaluated relative to this distribution, for example by assessing its percentile rank or by computing the proportion of randomly formed portfolios that achieve higher performance. By combining statistical inference between rule-based portfolios with simulation-based benchmarking against random portfolios, the evaluation framework provides a comprehensive assessment of whether systematic portfolio construction rules meaningfully alter investment outcomes in a market characterized by skewed and concentrated stock returns.

4 Theory

This chapter presents the theoretical concepts that provide the foundation for the empirical analysis conducted in this study. The purpose is to introduce the key financial and statistical concepts required to analyze the distributional properties of stock returns, the concentration of wealth creation across firms, and the evaluation of portfolio performance. The chapter begins by introducing common measures of asset returns, including arithmetic and logarithmic returns, which form the basis for empirical analysis of financial time series. These return measures are then considered in the context of return distributions, where particular attention is given to skewness as a measure of asymmetry. Understanding the distributional shape of stock returns is essential for analyzing the presence of extreme outcomes and the role they play in driving long-term investment results.

Building on the statistical properties of return distributions, the concept of wealth creation is introduced as a framework for measuring the economic contribution of individual firms to aggregate market outcomes. This framework allows the total value generated in the equity market to be decomposed into firm-level contributions and provides a way to analyze how concentrated long-term value creation is across companies.

Finally, the chapter presents the statistical tools used to evaluate portfolio performance in the empirical analysis. In particular, the paired t-test with Newey–West adjusted standard errors is introduced as a method for testing differences in average portfolio returns, and Monte Carlo simulation is presented as a framework for benchmarking rule-based portfolios against randomly formed portfolios. Together, these theoretical concepts provide the analytical foundation for the empirical investigation of skewness, wealth concentration, and portfolio outcomes in Nordic equity markets.

4.1 Arithmetic returns

Arithmetic returns represent one of the most commonly used measures of asset performance in financial analysis (Missiakoulis, Vasiliou, and Eriotis, 2008). The arithmetic return measures the simple percentage change in the price of an asset between two points in time and provides an intuitive measure of the gain or loss experienced by an investor over a given period.

Formally, the arithmetic return for asset i over period t is defined as

$$R_{i,t} = \frac{P_{i,t} - P_{i,t-1}}{P_{i,t-1}} \tag{4.1}$$

where $P_{i,t}$ denotes the asset price at time t and $P_{i,t-1}$ denotes the asset price in the previous period (Hudson and Gregoriou, 2010). When dividends or other cash flows are included, the return can instead be expressed as

$$R_{i,t} = \frac{P_{i,t} - P_{i,t-1} + D_{i,t}}{P_{i,t-1}} \tag{4.2}$$

where $D_{i,t}$ represents dividends received during the period.

Arithmetic returns are widely used in empirical finance due to their straightforward interpretation and their compatibility with many statistical methods. In particular, arithmetic returns are additive across assets within a single time period, which makes them suitable for calculating portfolio returns. For a portfolio consisting of n assets with weights w_i, the portfolio return in period t can be written as

$$R_{p,t} = \sum_{i=1}^{n} w_i R_{i,t} \tag{4.3}$$

This linear aggregation property makes arithmetic returns especially convenient in portfolio analysis and cross-sectional empirical studies.

4.2 Logarithmic returns

Logarithmic returns, often referred to as continuously compounded returns (Hudson and Gregoriou, 2010), represent an alternative way of measuring asset performance. Instead of measuring the simple percentage change in price, logarithmic returns measure the logarithmic change in asset prices between two points in time. This transformation provides several useful mathematical properties that make logarithmic returns widely used in financial modeling and econometrics.

The logarithmic return for asset i over period t is defined as

$$r_{i,t} = \ln\left(\frac{P_{i,t}}{P_{i,t-1}}\right) \tag{4.4}$$

where $P_{i,t}$ denotes the asset price at time t and $P_{i,t-1}$ denotes the price in the previous period (Hudson and Gregoriou, 2010). The logarithmic return can also be expressed in terms of the arithmetic return $R_{i,t}$ as

$$r_{i,t} = \ln(1 + R_{i,t}) \tag{4.5}$$

One important advantage of logarithmic returns is that they are additive across time. For a sequence of returns over T periods, the cumulative logarithmic return is simply the sum of the individual period returns

$$r_{i,(1,T)} = \sum_{t=1}^{T} r_{i,t} \tag{4.6}$$

This additive property simplifies the analysis of multi-period investment performance and therefore makes logarithmic returns particularly convenient in a financial setting (Hudson and Gregoriou, 2010).

4.3 Skewness

Skewness is a statistical measure that describes the asymmetry of a probability distribution around its mean. It provides information about how observations are distributed in the tails relative to the center of the distribution. In a perfectly symmetrical distribution, such as the normal distribution, skewness equals zero and the mean, median, and mode coincide. When a distribution is skewed, the mass of observations is unevenly distributed around the mean, which causes these central tendency measures to differ. A positively skewed distribution is characterized by a longer right tail, indicating that a small number of large positive observations pull the mean above the median. Conversely, a negatively skewed distribution has a longer left tail, where extreme negative observations pull the mean below the median.

The measure is defined as

$$\begin{aligned}
\tilde{\mu}_3 &= \mathbb{E}\left[\left(\frac{X-\mu}{\sigma}\right)^3\right] \\
&= \frac{\mathbb{E}[X^3] - 3\mu\,\mathbb{E}[X^2] + 3\mu^2\,\mathbb{E}[X] - \mu^3}{\sigma^3} \\
&= \frac{\mathbb{E}[X^3] - 3\mu\left(\mathbb{E}[X^2] - \mu\,\mathbb{E}[X]\right) - \mu^3}{\sigma^3} \\
&= \frac{\mathbb{E}[X^3] - 3\mu\sigma^2 - \mu^3}{\sigma^3}
\end{aligned}$$

Where X is a random variable, μ the mean and σ the standard deviation.

4.4 Wealth creation

An important implication of the asymmetric distribution of individual stock returns is that long-term value creation in equity markets tends to be highly concentrated among a small subset of firms. While many firms generate modest or negative long-run returns, a limited number of companies account for a large share of the total wealth created in the market. To analyze this phenomenon, Bessembinder (2018) introduces a framework for measuring firm-level wealth creation relative to a risk-free investment.

The concept of wealth creation measures the monetary value generated by a firm's equity in excess of the return that could have been earned by investing the same capital in a risk-free asset. Instead of focusing solely on percentage returns, this approach evaluates the dollar value added to investors over time. This distinction is important because firms differ substantially in size, and therefore a given percentage return may translate into very different levels of wealth creation depending on the firm's market capitalization.

Formally, the wealth created by firm i during period t can be expressed as

$$WC_{i,t} = MCAP_{i,t-1}\left(R_{i,t} - R_{f,t}\right), \tag{4.7}$$

where $MCAP_{i,t-1}$ denotes the market capitalization of firm i at the beginning of period t, $R_{i,t}$ represents the total return of the firm's equity during the period, and $R_{f,t}$ denotes the return on the risk-free asset. This formulation captures the excess value generated for investors during the period relative to the risk-free benchmark.

The total wealth created by a firm over its lifetime is obtained by aggregating these periodic contributions. In order to make contributions from different periods comparable, each period's wealth creation is typically expressed in terms of a common end-of-sample value by compounding with the risk-free rate. The cumulative wealth creation for firm i can therefore be written as

$$WC_i = \sum_{t=1}^{T} WC_{i,t} \prod_{s=t+1}^{T} (1 + R_{f,s}), \tag{4.8}$$

where T denotes the final observation period.

This framework highlights that total wealth creation depends not only on the returns generated by a firm, but also on the amount of capital invested in the firm over time. Large firms that generate moderate returns may therefore contribute more to total market wealth creation than smaller firms that experience extremely high percentage returns.

Aggregating wealth creation across firms allows for the measurement of total wealth generated in the equity market. The total market wealth creation can be expressed as

$$WC_{market} = \sum_{i=1}^{N} WC_i, \tag{4.9}$$

where N denotes the number of firms in the market and WC_i represents the cumulative wealth creation of firm i. This aggregation captures the total value created by the equity market in excess of the risk-free benchmark over the sample period.

By comparing each firm's wealth creation to the total market wealth creation, it is possible to evaluate the relative contribution of individual firms to aggregate market value creation. This provides a framework for analyzing the concentration of wealth creation across firms and identifying the extent to which a small subset of companies accounts for a disproportionate share of the total value generated in the market.

In the context of this study, the concept of wealth creation provides a framework for quantifying how value generated in Nordic equity markets is distributed across firms. By measuring firm-level contributions to aggregate market wealth, it becomes possible to evaluate the extent to which long-term value creation is concentrated among a small subset of companies.

4.5 Statistical t-test

The t-test is a statistical method used to determine whether there exists a statistically significant difference between two sample means. Several variants of the t-test exist depending on the structure of the data and the relationship between the samples. Common forms include the one-sample t-test, the independent samples t-test, and the paired samples t-test.

In the context of comparing portfolio construction methods, the paired samples t-test is the most appropriate. This is because the returns of the portfolios are observed over the same time periods, meaning that each observation forms a natural pair. Since all portfolios are exposed to the same market conditions at a given time, their returns are not independent. Market-wide movements tend to affect all portfolios simultaneously, creating correlation between the observations.

4.5.1 Hypothesis testing

Hypothesis testing provides a statistical framework for evaluating whether observed differences in sample data reflect genuine underlying effects or are likely to have occurred by random variation. In the context of this study, hypothesis testing is used to assess whether differences in returns between portfolios constructed according to different rules are statistically significant. The null hypothesis represents the absence of a systematic difference in performance, while the alternative hypothesis implies that one portfolio construction method yields a different average return than the other.

To conduct the test, the time series of return differences between two portfolios is defined as $d_t = R_t^A - R_t^B$. The hypothesis test then evaluates whether the expected value of these differences is equal to zero. If the null hypothesis is rejected, it indicates that the average return difference between the portfolios is statistically significant, suggesting that the portfolio construction methods lead to systematically different performance.

$$H_0 : \mathbb{E}[d_t] = 0, \qquad H_1 : \mathbb{E}[d_t] \neq 0$$

4.5.2 Newey-West

To assess whether differences in portfolio performance are statistically significant, mean return differences between portfolio strategies are evaluated using Newey–West adjusted t-statistics. The Newey–West estimator provides heteroskedasticity and autocorrelation consistent standard errors, which are appropriate for financial return series that typically exhibit time-varying volatility and serial correlation. This allows for robust inference when testing whether the average return difference between two portfolio construction methods is significantly different from zero.

Heteroskedasticity occurs when the variance of the residuals varies across observations or over time. In financial time series, this phenomenon often appears as volatility clustering, where periods of high volatility tend to be followed by further periods of high volatility, while calm periods are followed by relatively low volatility. In such cases, the assumption of constant variance is violated.

Another common feature of time series data is autocorrelation (or serial correlation), which arises when residuals are correlated across time. This implies that the error term in one period may be related to error terms in previous periods. As a consequence, the assumption that residuals are independent is no longer valid.

When both heteroskedasticity and autocorrelation are present, the variance–covariance matrix of the error terms is no longer diagonal. Instead, it contains time-varying variances on the diagonal and non-zero covariances between observations across time. Under these circumstances, conventional standard error estimators are inconsistent. To obtain reliable statistical inference, heteroskedasticity and autocorrelation consistent (HAC) estimators are required, of which the Newey–West estimator is the most widely used.

When the theory in this section and section 4.5 is combined the t-test can be constructed as follows.

Let R_t^A and R_t^B be observed returns for two portfolios constructed with different methods.

$$d_t = R_t^A - R_t^B \tag{4.10}$$

$$\bar{d} = \frac{1}{T}\sum_{t=1}^{T} d_t \tag{4.11}$$

To handle the presence of heteroskedasticity and auto-correlation the variance for $\bar{d}$ is calculated with Newey-West method. This gives the following formula for the t-statistic.

$$t = \frac{\bar{d}}{\sqrt{Var_{NW}(\bar{d})}} \tag{4.12}$$

To perform the test it is necessary to calculate the Newey-West standard error, which is the denominator in 4.12.

The Newey–West estimator adjusts the variance of the sample mean by incorporating weighted autocovariances of the return difference series. As described in Wooldridge (2013), the Newey–West variance estimator for $\bar{d}$ can be written as

$$Var_{NW}(\bar{d}) = \frac{1}{T}\left(\gamma_0 + 2\sum_{k=1}^{L} w_k \gamma_k\right) \tag{4.13}$$

where T denotes the number of observations, L is the chosen lag length, and w_k are lag-dependent weights defined as

$$w_k = 1 - \frac{k}{L+1}. \tag{4.14}$$

The term γ_k represents the sample autocovariance of the return difference series at lag k and is given by

$$\gamma_k = \frac{1}{T}\sum_{t=k+1}^{T} (d_t - \bar{d})(d_{t-k} - \bar{d}), \quad k = 1, 2, \ldots, L \tag{4.15}$$

while γ_0 corresponds to the sample variance of the return difference series

$$\gamma_0 = \frac{1}{T}\sum_{t=1}^{T} (d_t - \bar{d})^2. \tag{4.16}$$

The Newey–West standard error is then obtained as

$$SE_{NW}(\bar{d}) = \sqrt{Var_{NW}(\bar{d})}, \tag{4.17}$$

which is used in Equation 4.12 to compute the t-statistic.

4.6 Monte carlo simulation

5 Method

In this chapter, the method is described in a way which shows how the work was actually carried out. The description must be precise and well thought through. Consider the scientific term replicability. Replicability means that someone reading a scientific report should be able to follow the method description and then carry out the same study and check whether the results obtained are similar. Achieving replicability is not always relevant, but precision and clarity is.

Sometimes the work is separated into different parts, e.g. pre-study, implementation and evaluation. In such cases it is recommended that the method chapter is structured accordingly with suitable named sub-headings.

6 Results

This chapter presents the results. Note that the results are presented factually, striving for objectivity as far as possible. The results shall not be analyzed, discussed or evaluated. This is left for the discussion chapter.

In case the method chapter has been divided into subheadings such as pre-study, implementation and evaluation, the result chapter should have the same sub-headings. This gives a clear structure and makes the chapter easier to write.

In case results are presented from a process (e.g. an implementation process), the main decisions made during the process must be clearly presented and justified. Normally, alternative attempts, etc, have already been described in the theory chapter, making it possible to refer to it as part of the justification.

7 Discussion

This chapter contains the following sub-headings.

7.1 Results

Are there anything in the results that stand out and need be analyzed and commented on? How do the results relate to the material covered in the theory chapter? What does the theory imply about the meaning of the results? For example, what does it mean that a certain system got a certain numeric value in a usability evaluation; how good or bad is it? Is there something in the results that is unexpected based on the literature review, or is everything as one would theoretically expect?

7.2 Method

This is where the applied method is discussed and criticized. Taking a self-critical stance to the method used is an important part of the scientific approach.

A study is rarely perfect. There are almost always things one could have done differently if the study could be repeated or with extra resources. Go through the most important limitations with your method and discuss potential consequences for the results. Connect back to the method theory presented in the theory chapter. Refer explicitly to relevant sources.

The discussion shall also demonstrate an awareness of methodological concepts such as replicability, reliability, and validity. The concept of replicability has already been discussed in the Method chapter (5). Reliability is a term for whether one can expect to get the same results if a study is repeated with the same method. A study with a high degree of reliability has a large probability of leading to similar results if repeated. The concept of validity is, somewhat simplified, concerned with whether a performed measurement actually measures what one thinks is being measured. A study with a high degree of validity thus has a high level of credibility. A discussion of these concepts must be transferred to the actual context of the study.

The method discussion shall also contain a paragraph of source criticism. This is where the authors' point of view on the use and selection of sources is described.

In certain contexts it may be the case that the most relevant information for the study is not to be found in scientific literature but rather with individual software developers and open

source projects. It must then be clearly stated that efforts have been made to gain access to this information, e.g. by direct communication with developers and/or through discussion forums, etc. Efforts must also be made to indicate the lack of relevant research literature. The precise manner of such investigations must be clearly specified in a method section. The paragraph on source criticism must critically discuss these approaches.

Usually however, there are always relevant related research. If not about the actual research questions, there is certainly important information about the domain under study.

7.3 The work in a wider context

There must be a section discussing ethical and societal aspects related to the work. This is important for the authors to demonstrate a professional maturity and also for achieving the education goals. If the work, for some reason, completely lacks a connection to ethical or societal aspects this must be explicitly stated and justified in the section Delimitations in the introduction chapter.

In the discussion chapter, one must explicitly refer to sources relevant to the discussion.

8 Conclusion

This chapter contains a summarization of the purpose and the research questions. To what extent has the aim been achieved, and what are the answers to the research questions?

The consequences for the target audience (and possibly for researchers and practitioners) must also be described. There should be a section on future work where ideas for continued work are described. If the conclusion chapter contains such a section, the ideas described therein must be concrete and well thought through.

Bibliography

Bessembinder, Hendrik (2018). "Do Stocks Outperform Treasury Bills?" In: *Journal of Financial Economics*.

Campbell, John Y., Andrew W. Lo, and A. Craig MacKinlay (1997). *The Econometrics of Financial Markets*. Princeton University Press.

Cont, Rama (2001). "Empirical properties of asset returns: stylized facts and statistical issues". In: *Quantitative Finance* 1.2, pp. 223–236. DOI: 10.1080/14697680108916295.

Doane, David P. and Lori E. Seward (2011). "Measuring Skewness: A Forgotten Statistic?" In: *Journal of Statistics Education* 19.2. DOI: 10.1080/10691898.2011.11889611.

Financial Edge Training (2022). *Survivorship Bias*. https://www.fe.training/free-resources/asset-management/survivorship-bias/. Accessed: 2026-02-24.

Hudson, Robert S. and Andros Gregoriou (2010). *Calculating and Comparing Security Returns is Harder than You Think: A Comparison between Logarithmic and Simple Returns*. Tech. rep. SSRN Electronic Journal. URL: https://ssrn.com/abstract=1549328.

Miskolczi, Panna (2017). "Note on Simple and Logarithmic Return". In: *Applied Studies in Agribusiness and Commerce – APSTRACT* 11.1–2, pp. 127–136. ISSN: 1789-7874. DOI: 10.19041/APSTRACT/2017/1-2/16.

Missiakoulis, Spyros, Dimitrios Vasiliou, and Nikolaos Eriotis (2008). *Arithmetic Mean: A Bellwether for Unbiased Forecasting of Portfolio Performance*. Tech. rep. 1102194. SSRN. URL: https://ssrn.com/abstract=1102194.

Mota, Pedro P. (2012). "Normality assumption for the log-return of the stock prices". In: *Discussiones Mathematicae: Probability and Statistics* 32, pp. 47–58. DOI: 10.7151/dmps.1143.

Newey, Whitney K. and Kenneth D. West (1987). "A Simple, Positive Semi-Definite, Heteroskedasticity and Autocorrelation Consistent Covariance Matrix". In: *Econometrica* 55.3, pp. 703–708.

Reinholtz, Nicholas, Philip M. Fernbach, and Bart de Langhe (2016). *Do People Understand the Benefit of Diversification?* Working Paper. Available at SSRN: 2719144.

Wikimedia Commons contributors (n.d.). *Relationship between mean and median under different skewness*. https://commons.wikimedia.org/wiki/File:Relationship_between_mean_and_median_under_different_skewness.png. Accessed: 2026-03-03.

Wooldridge, Jeffrey M. (2013). *Introductory Econometrics: A Modern Approach*. 5th ed. South-Western Cengage Learning.